POSTMORTEM C

Forensic Medicine | 3rd Year MBBS

Hi Sammy!
PM changes = backbone
of forensic medicine!
Super high yield topic!

Postmortem Changes | Forensic Med | 3rd Year | MBBS

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PM changes help us estimate time since death. Learn the order!

DEFINITION

"Physical & chemical changes that occur in the body after death." — Reddy

Classification of Postmortem Changes

Category	Changes
Immediate	Insensibility, loss of voluntary movement, cessation of respiration & circulation
Early (0-72 hrs)	Algor mortis, livor mortis, rigor mortis, PM contact flattening
Late (>72 hrs)	Decomposition (putrefaction), adipocere, mummification

Algor Mortis (Cooling of the Body)

Feature	Details
Mechanism	Loss of hypothalamic thermoregulation after death
Rate of cooling	~1.5 F (0.8 C) per hour initially; then slower
Cooling curve	Sigmoid shaped — initial plateau, then linear fall, then plateau
Measurement	Rectal thermometer (0-50 C), inserted 10 cm, hourly readings
Formula	Time since death = (98.6 F - Rectal temp) / 1.5
Henssge nomogram	Most accurate modern method; uses rectal temp + body weight

Factors Affecting Algor Mortis

FASTER COOLING

- Thin / emaciated body
- Naked body / less clothing
- Cold environment / wind
- Infants and children (large SA:volume)

SLOWER COOLING

- Obese body (fat = insulator)
- Well-clothed body
- Hot, humid environment
- Febrile illness before death

★ Postmortem Caloricity!

Body temp RISES after death in: Septicemia, Tetanus, Strychnine poisoning, Pontine hemorrhage, Status epilepticus, Heat stroke. Mnemonic: STOPS Heat!

Livor mortis vs bruise and Nysten's rule are asked EVERY exam!

Postmortem Lividity (Livor Mortis / Hypostasis)

Feature	Details
Definition	Settling of blood in dependent parts due to gravity
Onset	30 min - 1 hour after death
Well developed	3-4 hours after death
Fixed	6-12 hours (no blanching on pressure)
Color	Bluish-purple (usual); pink (CO, cyanide, cold); brown (nitrites)
Absent in	Severe anemia, massive hemorrhage

PM Lividity vs Antemortem Bruise (VIVA FAVORITE!)

Feature	PM Lividity	Bruise
Location	Dependent parts only	Any part of body
Margins	Ill-defined, diffuse	Well-defined, localized
Incision	Blood washes away	Blood infiltrated (clotted)
Microscopy	Blood in vessels	Blood extravasated in tissues
Pressure	Blanches if not fixed	Does not blanch
Swelling	Absent	Present

Rigor Mortis (PM Stiffening)

Feature	Details
Cause	Depletion of ATP -> actin-myosin binding -> muscle stiffening
Onset	1-2 hours after death (small muscles first)
Fully developed	12 hours after death
Maintained	Next 12 hours (total 24 hrs)
Disappears	12-24 hours later (with onset of putrefaction)

★ **Nysten's Rule (Order of Rigor)**
Eyelids -> Jaw -> Neck -> Trunk -> Upper limbs -> Lower limbs
Rigor appears AND disappears in the same order! (Small -> Large muscles)

Special Types of Rigor

Type	Details
Cadaveric spasm	Instantaneous rigor at death — no primary flaccidity; indicates last act

Late PM changes are high yield for SAQs! Remember the timeline!

Putrefaction (Decomposition)

Timeline	Changes
24-36 hrs	Greenish discoloration of right iliac fossa (FIRST EXTERNAL SIGN)
36-48 hrs	Greenish discoloration spreads to abdomen, flanks
48-72 hrs	Marbling — hemolysis stains veins (purplish network)
3-5 days	Bloating (gas formation), purging of fluids from orifices
1-2 weeks	Maggot infestation, skin slippage, hair loosening
2-4 weeks	Tissue liquefaction, features unrecognizable
Months-Years	Skeletonization (soft tissue completely gone)

★ **KEY: First internal organ to putrefy = Larynx & Trachea (mucosa)**
Last internal organ = Uterus (non-gravid) / Prostate (male)
Casper's dictum: 1 week air = 2 weeks water = 8 weeks earth (decomposition ratio 1:2:8)

Adipocere (Saponification)

Feature	Details
Definition	Waxy transformation of subcutaneous fat after death
Conditions	Warm + moist + anaerobic (hot, humid, water immersion)
Organism	Clostridium welchii (anaerobic bacteria)
Appearance	Yellowish-white, waxy, greasy, sweet rancid smell
Timeline	Starts 3 weeks, complete in 3-6 months
Importance	Preserves body contour; retains injuries; helps TSD

Mummification

Feature	Details
Definition	Desiccation & shriveling of body due to dehydration
Conditions	Hot + dry + well-ventilated (desert, dry rooms)
Appearance	Dark brown, leathery, shrunken, hard, preserved
Timeline	Starts 3 months, complete in 6-12 months
Importance	Preserves injuries, features, identity possible

★ Adipocere vs Mummification!
Adipocere = WET (water) + WARM | Mummification = DRY + HOT
Both PRESERVE the body! Adipocere -> fat areas; Mummification -> entire body

TSD estimation = the MOST important skill in forensic medicine!

Time Since Death — Master Timeline Table

Time After Death	PM Change	Key Indicator
0-30 min	Primary flaccidity	Muscles relaxed, no changes yet
30 min-1 hr	Livor mortis begins	Faint discoloration, dependent parts
1-2 hrs	Rigor mortis starts	Eyelids and jaw first (Nysten's rule)
2-4 hrs	Algor mortis progressing	Body cooling ~1.5 F/hr
3-6 hrs	Lividity well-developed	Blanches on pressure (not yet fixed)
6-12 hrs	Lividity fixed	Does NOT blanch on pressure
12 hrs	Rigor fully developed	All muscles stiff
24-36 hrs	Putrefaction starts	Green discoloration, right iliac fossa
36-72 hrs	Rigor disappears	Secondary flaccidity; marbling appears
3-5 days	Bloating, gas formation	Distension, purging, foul smell
1-2 weeks	Maggot infestation	Skin slippage, hair loosening

Forensic Entomology (Insect Evidence)

Stage	Insect	Timeline / Significance
First to arrive	Blowflies (Calliphora)	Within minutes to hours of death
Egg -> Larva	Maggots (1st instar)	12-24 hours after egg laying
Active decay	Beetles (Dermestes)	Days to weeks; feed on dried tissue
Advanced decay	Piophila casei (cheese fly)	Weeks to months; indicates time
Skeletonization	Mites, beetles	Months to years

Other Methods of TSD Estimation

- **Vitreous humor K+** — Potassium rises linearly; useful up to 24-48 hrs after death
- **Gastric contents** — Stomach empty = 4-6 hrs after last meal; food type helps
- **CSF biochemistry** — Amino acids, enzymes increase with time
- **Bone changes** — Luminescence test, citrate content, nitrogen ratio

★ **Correct Sequence of PM Changes!**
Primary Flaccidity -> Rigor Mortis -> Secondary Flaccidity -> Marbling -> Putrefaction

PM changes = guaranteed
15+ marks in forensic!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

- 1. Describe postmortem changes with their medicolegal significance
- 2. Describe rigor mortis — definition, mechanism, stages, Nysten's rule, ML significance
- 3. Describe PM lividity. Differentiate from antemortem bruise (table!)
- 4. Describe methods of estimation of time since death in detail
- 5. Write about late PM changes — putrefaction, adipocere, mummification

Most Repeated SAQs (5–6 marks)

- 1. Nysten's rule and factors affecting rigor mortis
- 2. Cadaveric spasm — definition, examples, ML significance
- 3. Algor mortis — formula for TSD calculation
- 4. Casper's dictum (decomposition ratio in air:water:earth)
- 5. Adipocere — formation, conditions, significance
- 6. Differentiate PM lividity from bruise (table format!)

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
First external sign of putrefaction	Green discoloration, right iliac fossa
First internal organ to putrefy	Larynx & Trachea (mucosa)
Last organ to putrefy (female)	Non-gravid uterus
Last organ to putrefy (male)	Prostate
Instantaneous rigor without flaccidity	Cadaveric spasm
Pugilistic/Boxer's attitude	Heat stiffening (>65 C)
Goose skin after death	Cutis anserina (arrector pili rigor)
Pink PM lividity	CO poisoning / Cyanide / Cold
Cherry red lividity	Carbon monoxide poisoning
Brown PM lividity	Nitrite / aniline poisoning
Casper's dictum ratio	Air : Water : Earth = 1 : 2 : 8
Organism for adipocere	Clostridium welchii

High Yield Checklist

MUST KNOW

- Nysten's rule
- Lividity vs Bruise
- Algor mortis formula
- Cadaveric spasm
- Putrefaction timeline

HIGH YIELD

- Casper's dictum
- Adipocere conditions
- PM caloricity causes
- Heat stiffening
- TSD methods (vitreous K+)

BONUS

- Forensic entomology
- Henssge nomogram
- Mummification conditions
- Tardieu spots
- Gastric contents TSD

♥ Every PM finding tells a story, Sammy! You'll speak for the dead ♥